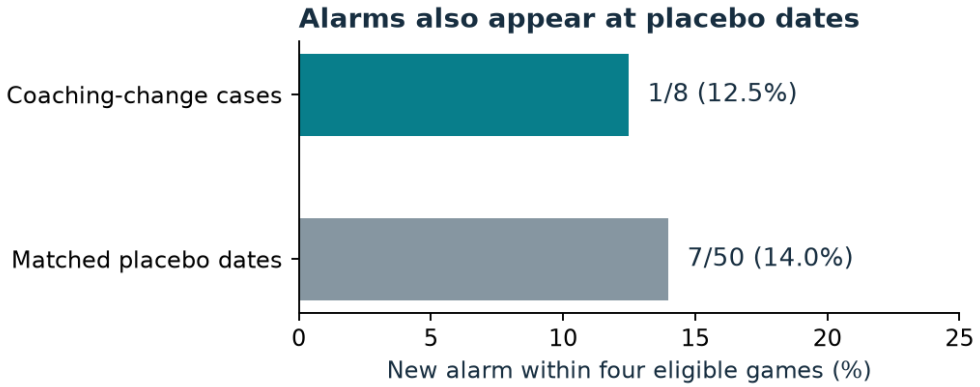
Drives as a Production Line

Did offensive efficiency shift after a mid-season coaching change?

1 of 8 eligible coaching changes produced a new control-chart alarm within four games.

The identical procedure flagged **7 of 50 matched placebo dates (14%)**. This small observational sample does not show a clear concentration of alarms after the selected changes.



What the number means

The new alarm belongs to Nebraska in 2022 and is a downward WECO signal. Charlotte had an alarm before its change. Texas A&M and Boise State have only one and three eligible follow-up games, respectively, so neither enters the four-game comparison. No first CUSUM crossing landed within one eligible game of a change.

Method

Ten documented coaching changes; fixed prior-season baselines of 11-12 games. Opponent and home-field effects are fitted only on the prior season, excluding the focal offense from training. Per-game adjusted PPA is monitored with I-MR charts, four Western Electric rules and a date-blind two-sided CUSUM (k=0.5, h=5). A new alarm must first arise in the four-game follow-up, with no earlier target-season alarm.

Why caution changes the conclusion

The Nebraska alarm disappears when baseline sigma uses the sample standard deviation instead of moving ranges. The placebo rate has an approximate 95% Wilson interval of 7%-26%; controls are not known to be change-free. Four games offer limited precision, and coaching changes coincide with injuries, staff changes and schedule shifts. An alarm is not proof of a coaching effect; no alarm is not proof of no effect.

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Evidence, data quality and practical interpretation

All ten cases remain visible

Team / season	Post games	Primary classification	Delta PPA*
Nebraska 2022	9	new post-change alarm	-0.128
Arizona State 2022	9	no new alarm	-0.140
Georgia Tech 2022	8	no new alarm	-0.104
Colorado 2022	7	no new alarm	-0.129
Wisconsin 2022	7	no new alarm	+0.135
Charlotte 2022	4	preexisting alarm	+0.129
Auburn 2022	4	no new alarm	-0.014
Michigan State 2023	10	no new alarm	-0.218
Texas A&M 2023	1	insufficient follow-up	Not assessed
Boise State 2023	3	insufficient follow-up	Not assessed

*First-four-game mean minus prior-season adjusted baseline; descriptive, not causal. Bootstrap intervals and all sensitivity variants are in the companion tables. Boise State includes the conference championship classified as regular season by CFBD.

Coverage and exclusions

634,546 raw plays; 285,315 eligible offensive plays; 284,804 with PPA. All 2,216 completed FBS-vs-FBS games have records. Regulation only; kneels, spikes, penalties, special teams and fourth-quarter margins above 28 are excluded. Drive analysis retains 48,692 drives after clock/outcome checks; 78 ambiguous drive IDs are excluded. Complete game coverage does not guarantee every play is recorded correctly.

Secondary result: tempo

An additional eligible snap per possession minute is associated with -0.039 points per drive (95% interval -0.127 to +0.049), controlling for offense, defense, season, starting field position and home field. The interval includes zero. This is game-clock pace, not real-time tempo. Points count offensive TDs as six and field goals as three; conversions are excluded.

Sources and reproducibility

Data provided by CollegeFootballData.com. Signal rules follow the NIST control-chart guidance and CUSUM guidance. Dated intervention sources, exclusions and limitations appear in interventions.csv and PROTOCOL.md. Scripts, derived tables, an analysis notebook and an offline dashboard accompany this report. The analysis plan was recorded after inspecting the Texas A&M pilot; it is not a preregistration.